

SMART INDIA HACKATHON 2026

TOP 10 PROBLEM STATEMENTS — BEGINNER FRIENDLY GUIDE

Bilkul Aasaan Bhasha Me Samjho: Problem, Solution, Tech Stack & Judges Presentation Tips

Bina Kisi Confusion Ke Shortlist Kaise Ho? (Winning Strategy)

Agar aap pehli baar SIH me participate kar rahe hain, to generic social/web apps me hazaro teams se compete mat karo. Top Ministries (ISRO, DRDO, MoES, Land Dept, Coal India) ke in **Top 10 Problem Statements** me se ek chuno. Inme:

- **Competition 90% Kam Hota Hai:** Mass spamming teams in technical problems se dur rehti hain.
- **Working Demo Dikahana Aasaan Hai:** Live Maps, AI Scanners, ya Visual Charts dekhte hi judges full marks dete hain.

RANK 1 — PS ID: 26014 | Integrated GIS Digital Public Infrastructure for Land Governance (Land Stack)

Ministry / Organization: Ministry of Rural Development

Shortlist Hone Ka Reason: Sarkar ka direct National Priority project hai. GIS map wala demo judges ko bahut impressed karta hai aur competitors kam hote hain.

1. Aasaan Samjh (What is the Problem?):

Zameen ke sabhi kaagaz (registry, tax, ownership, map) alag-alag offices me bikhre hain. Aapko ek single map-based website (Land Stack) banana hai jahan zameen par click karte hi uski saari jankari dikh jaye.

2. Aapko Kya Banana Hai (What to Build?):

Ek Google Maps jaisa dashboard jisme har zameen ka plot highlighted ho, aur click karne par owner name, land area, registry history, aur tax status dikhe.

3. Beginners Tech Stack (Kaise Banayein?):

Frontend: Leaflet.js / Mapbox + React | Backend: Python FastAPI | Database: PostGIS / PostgreSQL

Judges Ke Saamne Presentation Tip:

PPT me ek working map dikhao jisme 1 plot click karke owner details load ho rahi hon.

RANK 2 — PS ID: 26016 | Real-Time National Land Acquisition & Management System

Ministry / Organization: Ministry of Rural Development

Shortlist Hone Ka Reason: Multi-ministerial infrastructure tracking project hai. Workflow dashboard aur visual charts dekhte hi clear ho jate hain.

1. Aasaan Samjh (What is the Problem?):

Highways ya Railways ke liye zameen khareedne (land acquisition) ka process paper par chalta hai jisse saalo lag jate hain. Aapko iska complete digital monitoring system banana hai.

2. Aapko Kya Banana Hai (What to Build?):

Ek Web Portal jahan project proposals upload ho sakein, zameen ki geo-tagging dikhe, compensation (muavza) kitna diya gaya uska real-time bar-chart dikhe.

3. Beginners Tech Stack (Kaise Banayein?):

Frontend: React / HTML-CSS | Backend: Python Flask / Node.js | Database: MySQL / PostgreSQL

Judges Ke Saamne Presentation Tip:

Dashboard me Status Tracker dikhao: Proposal -> Notified -> Acquired -> Compensation Paid.

RANK 3 — PS ID: 26019 | National Digital Platform for Research & Policy Innovation in Land Governance

Ministry / Organization: Ministry of Rural Development

Shortlist Hone Ka Reason: AI + Knowledge Base ka unique combination hai. Easy to build, visual, aur innovative hai.

1. Aasaan Samjh (What is the Problem?):

Land laws aur research papers aam logo aur researchers ko nahi milte. Aapko ek AI-powered Knowledge Repository Portal banana hai.

2. Aapko Kya Banana Hai (What to Build?):

Ek Smart Research Search Engine portal jisme land policies, court judgments, aur GIS trends ka AI summary aur download option ho.

3. Beginners Tech Stack (Kaise Banayein?):

Frontend: React / Streamlit | Backend: Python (FastAPI) | AI Search: LangChain / Sentence-Transformers

Judges Ke Saamne Presentation Tip:

Search bar me "Land acquisition policy Punjab" type karke AI Summary aur PDF download demo dikhao.

RANK 4 — PS ID: 26024 | AI-Based Smart Governance & Compliance Monitoring for Coal Mines

Ministry / Organization: Ministry of Coal (Coal India / CMPDI)

Shortlist Hone Ka Reason: Coal India ka official problem hai. AI Helmet/Safety Detection ka live camera demo judges ko 100% impress karega.

1. Aasaan Samjh (What is the Problem?):

Koyla khano (Coal mines) me safety aur environment rules ki report manual banti hai jisse chori aur accidents hote hain. Isko AI se automate karna hai.

2. Aapko Kya Banana Hai (What to Build?):

Ek Central Web Dashboard + Field Mobile App. Inspector safety photo upload karega, AI check karega ki worker ne helmet pehna hai ya nahi, aur violations par auto-alert bhejega.

3. Beginners Tech Stack (Kaise Banayein?):

AI: Python OpenCV + YOLOv8 | Web: React / Streamlit | Mobile: Flutter / Progressive Web App

Judges Ke Saamne Presentation Tip:

Live webcam demo me helmet detection ka box dikhao aur Safety Compliance Score 95% generate karke dikhao.

RANK 5 — PS ID: 26057 | AI-Powered Underwater Marine Debris & Anomaly Detection System

Ministry / Organization: Ministry of Earth Sciences (MoES / NIOT)

Shortlist Hone Ka Reason: Deep-Tech Marine project hai. Normal students samundar ke projects nahi chunte, isliye competition lagbhag 0 hota hai!

1. Aasaan Samjh (What is the Problem?):

Samundar ke andar kharab machhli pakadne ke jaal (ghost nets) aur kachra underwater life ko khatam karte hain.

Sonar images se kachra pehchanne wala AI banana hai.

2. Aapko Kya Banana Hai (What to Build?):

Ek Web Application jahan Side-Scan Sonar (underwater image) upload karne par AI model kachre/jaal ke charo taraf red box (bounding box) draw kar de aur GPS location report kare.

3. Beginners Tech Stack (Kaise Banayein?):

AI: Python PyTorch / YOLOv8 / OpenCV | Web UI: Streamlit / Gradio / React

Judges Ke Saamne Presentation Tip:

Sonar image upload karke "Ghost Net Detected - Confidence 94% - Lat: 13.08, Long: 80.27" dikhao.

RANK 6 — PS ID: 26078 | AI-Driven Spatio-Temporal Tracking of Extreme Weather Anomalies

Ministry / Organization: Ministry of Earth Sciences (NCMRWF)

Shortlist Hone Ka Reason: Weather forecasting disaster management ka sabse hot topic hai. Charts aur Heatmaps evaluators ko bohot pasand aate hain.

1. Aasaan Samjh (What is the Problem?):

Toofan, bhayankar barish, aur heatwaves ko 3-10 din pehle track karna mushkil hota hai. Aapko weather data se extreme weather ko predict karne wala AI dashboard banana hai.

2. Aapko Kya Banana Hai (What to Build?):

Ek Weather Alert Map Dashboard jo rain/temperature data ko process karke bataye ki kis 5km area me next 24 hours me cloudburst ya heatwave ka risk hai.

3. Beginners Tech Stack (Kaise Banayein?):

Data/AI: Python Pandas + Scikit-Learn / XGBoost | Map UI: Leaflet.js / Streamlit

Judges Ke Saamne Presentation Tip:

Map par Red Heatmap Zone dikhao aur auto SMS Alert "High Flood Risk in 5km Radius" trigger karke dikhao.

RANK 7 — PS ID: 26094 | AI-Powered Mental Health Monitoring & Distress Prediction for Victims

Ministry / Organization: Ministry of Social Justice & Empowerment (MoSJE)

Shortlist Hone Ka Reason: Social cause + HealthTech hai. Emotional AI demo evaluators ke dil ko chhoo jata hai.

1. Aasaan Samjh (What is the Problem?):

Crime aur atrocities ke shikar logo me depression aur mental trauma ho jata hai. Sarkar unke mental well-being ko track karke help karna chahti hai.

2. Aapko Kya Banana Hai (What to Build?):

Ek AI Chatbot / Voice Assistant jo victim se simple baatein kare (Hindi/English me), unke jawab aur sentiment se Distress Level (Low/Medium/High) calculate kare aur counsellors ko alert kare.

3. Beginners Tech Stack (Kaise Banayein?):

NLP/AI: Python NLTK / VADER Sentiment / HuggingFace | Web Chatbot: React / Web Speech API

Judges Ke Saamne Presentation Tip:

Chatbot me victim ka response "Mujhe darr lag raha hai" daal kar "Distress Score: 85% (High Risk) - Counsellor Alerted" dikhao.

RANK 8 — PS ID: 26011 | 3D ULPIN Generation & Vertical Property Mapping System**Ministry / Organization:** Ministry of Rural Development (DoLR)**Shortlist Hone Ka Reason:** 3D Web visualization bohot rare hoti hai hackathons me. 3D building model ghumakar dikhane par 100% shortlist chance hai.**1. Aasaan Samjh (What is the Problem?):**

2D map par multi-storey building ke 20 flats ek hi zameen par dikhte hain. Aapko har flat aur basement ka alag 3D Box (3D ULPIN ID) wala map banana hai.

2. Aapko Kya Banana Hai (What to Build?):

Ek 3D Web Viewer jisme multi-storey building 3D me dikhe. Jab user Floor 5, Flat 502 par click kare to us flat ka 3D block highlight ho aur owner details aaye.

3. Beginners Tech Stack (Kaise Banayein?):

3D Web UI: Three.js / Cesium.js / Babylon.js | Backend: Python FastAPI / Node.js

Judges Ke Saamne Presentation Tip:

Browser me 3D Building ghumakar Flat 301 click karke uski ownership window khol kar dikhao.

RANK 9 — PS ID: 26012 | AI-Based Automated Urban Parcel Mapping using Drone Imagery**Ministry / Organization:** Ministry of Rural Development (DoLR)**Shortlist Hone Ka Reason:** Drone + AI fast cadastral mapping survey efficiency ko 100x badha deta hai. Direct high priority tool.**1. Aasaan Samjh (What is the Problem?):**

Drone se li gayi aerial photos se ghar aur sadkon ke border manually draw karne me hafte lagte hain. AI se borders auto-detect karne hain.

2. Aapko Kya Banana Hai (What to Build?):

Ek Web Tool jahan Drone Photo upload hoti hai, AI background me gharon ke charo taraf yellow boundaries auto-draw kar deta hai aur CAD/GIS file download karne deta hai.

3. Beginners Tech Stack (Kaise Banayein?):

AI: Python OpenCV + Segment Anything Model (SAM) / YOLOv8 | Web: Streamlit / Flask

Judges Ke Saamne Presentation Tip:

Raw Drone Image -> Click "Auto Segment" -> Yellow Parcel Lines Drawn instantly.

RANK 10 — PS ID: 26018 | Intelligent Land Record Digitization and Validation System (OCR)**Ministry / Organization:** Ministry of Rural Development (DoLR)**Shortlist Hone Ka Reason:** Har government department me record digitization ki zarurat hai. Live scanning demo se judges instant marks dete hain.**1. Aasaan Samjh (What is the Problem?):**

50 saal purane zameen ke handwritten kaagaz aur Urdu/Hindi registers corrupt ho rahe hain. Unse text extract karke computer database me feed karna hai.

2. Aapko Kya Banana Hai (What to Build?):

Ek Document Scanner Web App. Purane paper ki photo upload karte hi OCR name, khasra number, aur land area pehchan kar form fields me fill kar de.

3. Beginners Tech Stack (Kaise Banayein?):

OCR/AI: Tesseract OCR / EasyOCR / Python OpenCV | Web UI: React / Streamlit

Judges Ke Saamne Presentation Tip:

Purana handwritten kaagaz upload karo -> Screen par Name: "Ramesh Kumar", Khasra: "452/1" auto-fill hoke dikhao.